

**Course Code : ECT 351**

**GVHW/RW – 23 / 1213**

**Fifth Semester B. Tech. (Electronics and Communication Engineering)  
Examination**

**ELECTROMAGNETIC WAVES**

Time : 3 Hours]

[Max. Marks : 60

**Instructions to Candidates :—**

- (1) All questions are compulsory.
- (2) All questions carry marks as indicated.
- (3) Illustrate your answer with neat sketches.
- (4) Assume suitable data wherever necessary.

1.
  - (a) What are standing waves ? Why do we get standing waves on transmission line ? Also explain voltage standing wave ratio. 5(CO1)
  - (b) An impedance of  $80 - j30 \Omega$  is connected to  $50 \Omega$  line. Find the complex reflection coefficient at the load, the complex reflection coefficient and impedance at a distance of  $0.3 \lambda$  from the load. What is the VSWR measured on the line ? 5(CO1)
2.
  - (a) How does phase constant vary with frequency for a uniform plane wave ? 5(CO2)
  - (b) The magnetic field at the surface of a good conductor is  $2 \text{ A/m}$ . The frequency of the field is  $600 \text{ MHz}$ . If the conductivity of the conductor is  $10 \text{ mho/m}$ , find the skin depth of the conductor. 5(CO2)
3.
  - (a) Explain the behavior of plain waves at conducting interface. 5(CO4)
  - (b) a light beam is incident from air to a medium with a dielectric constant 5 and relative permeability 200. If the angle of incidence is  $50^\circ$ , find the angle of reflection and angle of refraction. 5(CO4)

4. (a) What is cut off frequency ? Formulate the equations for cut off frequency of TE waves in parallel plane waveguide. Show that the waveguide behaves like a high pass filter. 5(CO5)
- (b) An EM wave is propagating between parallel conducting planes in z direction, separated by 9.5 cm. Find the cut off frequency of  $TE_{10}$ ,  $TE_{20}$  and  $TE_{30}$  modes. Assume air as dielectric. 5(CO5)
5. (a) Derive the relationship between guide wavelength, free space wavelength and cut off wavelength for rectangular wave guides. 4(CO5)
- (b) What transmission modes are possible at an operating frequency of 3 GHz in rectangular waveguide with dimension 7.22 cm x 3.44 cm ? 6(CO5)
6. (a) What is the difference between directivity and directive gain of an antenna ? What is the effective aperture of an antenna ? 5(CO3)
- (b) Transmitting and receiving antennas operating at 5 GHz with gains of 30 dB and 25 dB respectively are separated by a distance of 5 km. Assume that both antennas are identical in polarization. Estimate the received power if the transmitted power is 200 W. 5(CO3)

